

Price Norm and Monetary Policy

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Real interest rate . key price in the model

↳ service. price of government bonds relative to
 $\dot{p}(t)/p(t)$, given by price nom

$$r(t) = \underset{\substack{\uparrow \\ \text{monetary policy}}}{i(t)} - \pi(t)$$

Price nom: prices grow at constant inflation rate π .

$$p(t) = p(0) \cdot \exp(\pi \cdot t)$$

$$\pi(t) = \frac{\dot{p}(t)}{p(t)} = \pi$$

Monetary policy: - central bank follows interest rate

peg

$$i(t) = i$$

- central bank is subject to

zero-lower-bound constraint: $i \geq 0$.
(ZLB)

Real interest is fixed

$$r = i - \pi$$

Assumption: $r < 0$, but r can be < 0